

April 2023

Duration: 3 Hours

Max. Marks: 100

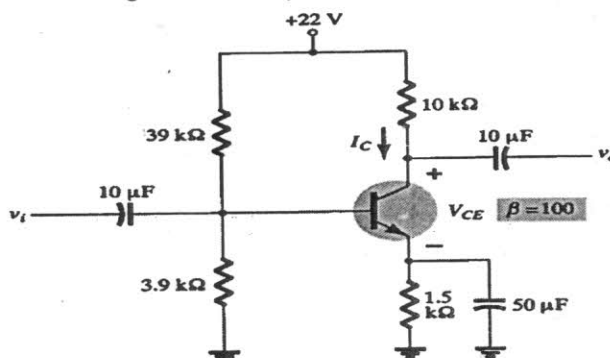
Unit – I

Marks	BT*	CO*	PO*
-------	-----	-----	-----

-

- b) For the voltage divider circuit with R_E bypassed $V_{CC}=12V$, $R_1=25k\Omega$, $R_2=4.5k\Omega$, $R_E=220\Omega$, $R_C=1.5k\Omega$, $C_{C1}=C_{C2}=10\mu F$, $C_E=100\mu F$, $h_{ie}=1.173k\Omega$, $h_{oe}=30\mu A/V$ and $h_{fe}=100$ find Z_i , Z_o , A_v and A_i .
 - c) Explain early voltage (V_A) and increase in output impedance (r_o) of a transistor with the V_I curve, r_e model and necessary equations.
- a) Draw the r_e model and derive expressions for Z_i , Z_o , A_v and A_i for emitter follower.
 - b) Determine the dc bias voltage V_{CE} and the current I_C for the following circuit using exact analysis.

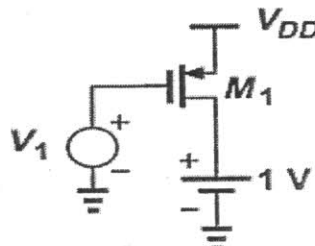
7	L*3	1	1
7	L3	1	1
6	L2	1	1
7	L2	1	1



- c) For emitter bypassed voltage divider bias circuit, derive the expression for the input impedance, output impedance and voltage gain using r_e model of transistor.

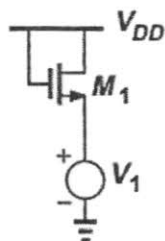
6	L3	1	1
7	L2	1	1

3. a) In the circuit below, determine the region of operation of M1 as V_1 goes from V_{DD} to zero. Assume $V_{DD} = 2.5\text{ V}$ and $|V_{TH}| = 0.5\text{ V}$



6 L3 2 1

- b) Sketch the drain current of M_1 versus V_1 as V_1 varies from zero to V_{DD} , justify the answer with MOS models and necessary I_D equation. Assume $\lambda = 0$.



- | | | | | | |
|-------------------|--|---|----|---|---|
| | | 7 | L3 | 2 | 1 |
| | c) With conceptual setup for measurement of f_T of MOS transistor explain transit frequency with necessary equations. | 7 | L2 | 2 | 1 |
| 4. | a) With the circuit diagram and expressions, explain MOS current mirror. | 7 | L2 | 3 | 1 |
| | b) Design a source follower to drive a 50Ω load with a voltage gain of 0.5 and a power budget of 10 mW. Assume $\mu_n C_{ox} = 100 \mu A/V^2$, $V_{TH} = 0.5 V$, $\lambda = 0$ and $V_{DD} = 1.8 V$ | 7 | L3 | 2 | 1 |
| | c) Obtain the voltage gain equation for CS amplifier with source degeneration with the help of circuit diagram and small signal model ($\lambda=0$). | 6 | L2 | 2 | 1 |
| 5. | a) Explain differential amplifier with resistive loads. | 7 | L2 | 3 | 1 |
| | b) Explain differential amplifier with active load. | 7 | L2 | 3 | 1 |
| | c) Explain CMRR w.r.to MOS pair with asymmetric loads. | 6 | L2 | 3 | 1 |
| Unit – III | | | | | |
| 6. | a) What is an instrumentation amplifier? What are its applications? With a neat circuit diagram explain an instrumentation amplifier using a transducer bridge. | 6 | L2 | 4 | 1 |
| | b) A Schmitt trigger circuit has $R_1 = 560 \Omega$, $R_2 = 68 k\Omega$, $V_{in} = 500 mV$ p-p sine wave and saturation voltages $\pm 14V$. Determine the threshold voltages V_{ut} and V_{lt} . Draw the circuit and waveforms. | 7 | L3 | 4 | 1 |
| | c) Explain the operation of 4-bit R-2R DAC with a neat circuit diagram and derive the expression for the output voltage. For a DAC with $R = 10 k\Omega$, $R_F = 20 k\Omega$ and $V_{REF} = 5 V$, determine the output voltage when the inputs are $b_0 = b_1 = 0 V$ and $b_2 = b_3 = 5 V$. | 7 | L3 | 5 | 1 |
| 7. | a) With a neat diagram, explain the working of Successive Approximation ADC. Illustrate the conversion of analog input $V_{in} = 6.2 V$ to an equivalent 4-bit binary value, assuming full-scale output of 15 V. | 6 | L3 | 5 | 1 |
| | b) Draw the circuit and frequency response of a first order high-pass filter using opamp and derive the expression for the magnitude of gain. Design a first order HPF to have a cut-off frequency of 2 kHz with a pass band gain of 2, choosing $C = 0.01 \mu F$. | 7 | L3 | 4 | 1 |
| | c) Design an astable multivibrator circuit using 555-timer, to get an output frequency of 5 kHz with 60% duty cycle, choosing $C = 0.068 \mu F$. Draw the circuit diagram and waveforms. | 7 | L3 | 5 | 1 |
| 8. | a) Explain precision full wave rectifier with the help of Block diagram, circuit and waveform. | 7 | L2 | 4 | 1 |
| | b) Draw the detailed circuit diagram of sample and hold circuit and explain the operation with necessary waveform. | 6 | L2 | 4 | 1 |
| | c) Design a mono-stable multi-vibrator using 555 timer to obtain a pulse of width 10msec where $R=10 k\Omega$. | 7 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
